LABORATORY TITLE: SMTP EMAIL TRAFFIC FORENSIC

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STUDENT ID: 2025/FWSD/11435

COURSE NAME: BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS

COURSE CODE: SBT-DF203 (LAB 4)

INSTRUCTOR NAME: MR. AMINU IDRIS

DATE OF SUBMISSION: SEPTEMBER 15TH, 2026.

VERSION: Version 1.0

Executive Summary

This report presents a forensic investigation of unencrypted Simple Mail Transfer Protocol (SMTP) traffic captured in smtp.pcap. Analysis reveals that client 10.10.1.4 established a session with mail server 74.53.140.153 (xc90.websitewelcome.com).

Although the server advertised STARTTLS capability in its initial EHLO response, the client never issued a STARTTLS command. Consequently, the entire interaction including user authentication (AUTH LOGIN) and multi-part email payload transmission occurred in cleartext. Base64-encoded credentials were extracted, decoded offline, and masked per reporting requirements. The investigation confirmed successful email delivery under queue ID 1Mugho-0003Dg-Un.

This investigation was authorized as part of SBT-DF203-Lab4. All forensic analysis was conducted on a read-only duplicate working copy (working/smtp_working.pcap) to preserve original evidence integrity.

SMTP Timeline

The session spans from 2009-10-05 02:06:08.219663 to 2009-10-05 02:06:15.105467 EDT (total duration: ~6.88 seconds).

Command/Response Analysis

Command and Response Sequence Table

Frame	Timestamp	Source IP	Destination IP	Command/Response Code	Parameter/Message Text
6	02:06:08.219663	74.53.140.153	10.10.1.4	220	xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009...
7	02:06:08.224809	10.10.1.4	74.53.140.153	EHLO	GP
9	02:06:08.566183	74.53.140.153	10.10.1.4	250	Hello GP [122.162.143.157] | SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP
10	02:06:08.568729	10.10.1.4	74.53.140.153	AUTH	LOGIN
11	02:06:08.911081	74.53.140.153	10.10.1.4	334	VXNlcm5hbWU6 *(Base64 for "Username:")*
12	02:06:08.911655	10.10.1.4	74.53.140.153	(User Param)	Z3VycGF0YXBAcGF0cmlvdHMuaW4=
13	02:06:09.253544	74.53.140.153	10.10.1.4	334	UGFzc3dvcmQ6 *(Base64 for "Password:")*
14	02:06:09.254118	10.10.1.4	74.53.140.153	(Pass Param)	cHVuamFiRGlQREyMw==
15	02:06:09.613798	74.53.140.153	10.10.1.4	235	Authentication succeeded
16	02:06:09.614414	10.10.1.4	74.53.140.153	MAIL FROM	<gurpartap@patriots.in>
17	02:06:09.956765	74.53.140.153	10.10.1.4	250	OK
18	02:06:09.957250	10.10.1.4	74.53.140.153	RCPT TO	<raj_deol2002in@yahoo.co.in>

19	02:06:10.319708	74.53.140.153	10.10.1.4	250	Accepted
20	02:06:10.320203	10.10.1.4	74.53.140.153	DATA	Initiate message data transmission
21	02:06:10.661679	74.53.140.153	10.10.1.4	354	Enter message, ending with "." on a line by itself
22-4	02:06:10.692743	10.10.1.4	74.53.140.153	DATA	Segmented message body fragments
52	02:06:12.248789	74.53.140.153	10.10.1.4	250	OK id=1Mugho-0003Dg-Un
54	02:06:14.763825	10.10.1.4	74.53.140.153	QUIT	Session termination request
56	02:06:15.105467	74.53.140.153	10.10.1.4	221	xc90.websitewelcome.com closing connection

Authentication Evidence and Redaction Method

Authentication was executed via AUTH LOGIN (cleartext challenge-response encoded in Base64).

Captured Raw Values

- Encoded Username (Frame 12): Z3VycGF0YXBAcGF0cmlvdHMuaW4=
- Encoded Password (Frame 14): cHVuamFiRGlQREyMw==

Redaction Method and Masked Output

Per forensic reporting standards, credentials decoded offline using Python are masked in public documentation:

- Public Masked Username: g*********p@patriots.in
- Public Masked Password: p*********w

Email Reconstruction

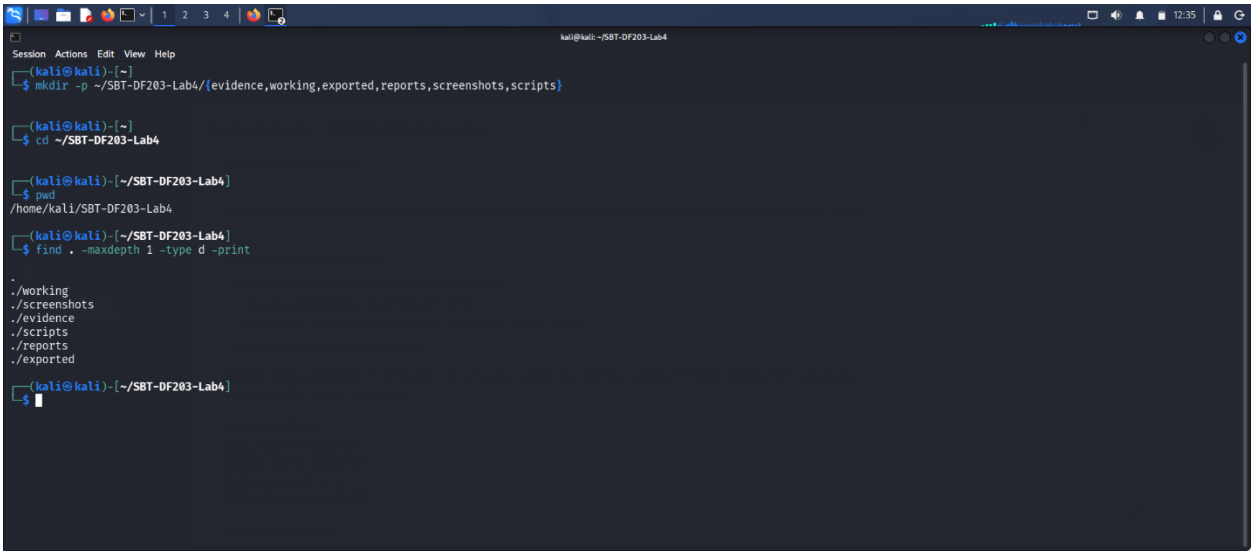
Envelope Parameters vs. RFC 2822 Headers

- Envelope Sender (MAIL FROM): gurpartap@patriots.in
- Envelope Recipient (RCPT TO): raj_deol2002in@yahoo.co.in
- Header From: "Gurpartap Singh" <gurpartap@patriots.in>
- Header To: raj_deol2002in@yahoo.co.in
- Subject: SMTP
- MIME Structure: Multipart message containing text/plain and text/html sub-parts.
- Attachment Present: No discrete file attachments were transferred; content consists of inline text and HTML formatting.

Client/Server and Network Metadata

- Client Endpoint: 10.10.1.4 (External NAT IP indicated in EHLO: 122.162.143.157)
- Server Endpoint: 74.53.140.153 (xc90.websitewelcome.com)
- Target Service Port: TCP Port 25 (Standard unencrypted SMTP port)
- Mail Server Banner: Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
- Observed Network Anomaly: ICMP Destination Unreachable (Fragmentation needed) packets (Frames 26–30) issued by router 192.168.1.1 during heavy DATA transmission, causing minor TCP out-of-order retransmissions (Frames 32–36).

Lab Folder Structure and Evidence Preparation





```
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab  evidence/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab  working/smtp_working.pcap

┌──(kali㉿kali)-[~/SBT-DF203-Lab4]
└─$ capinfos evidence/smtp.pcap | tee reports/smtp_capinfos.txt
File name:           evidence/smtp.pcap
File type:           Wireshark/tcpdump/... - pcap
File encapsulation:  Ethernet
File timestamp precision:  microseconds (6)
Packet size limit:   file hdr: 65535 bytes
Number of packets:   60
File size:           27 kB
Data size:           26 kB
Capture duration:    9.198384 seconds
Earliest packet time: 2009-10-05 02:06:07.492060
Latest packet time:   2009-10-05 02:06:16.690444
Data byte rate:      2,920 bytes/s
Data bit rate:       23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256:              17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1:                3def1a1fed849e7f23b66e6925ddff05845a400b
Strict time order:   True
Number of interfaces in file: 1
Interface #0 info:
                     Encapsulation = Ethernet (1 - ether)
                     Capture length = 65535
                     Time precision = microseconds (6)
                     Time ticks per second = 1000000
                     Number of stat entries = 0
                     Number of packets = 60

┌──(kali㉿kali)-[~/SBT-DF203-Lab4]
└─$
```

Part A - Inventory the Capture and Locate SMTP Streams

```
┌──(kali㉿kali)-[~/SBT-DF203-Lab4]
└─$ tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt
================================================================================
TCP Conversations
Filter:<No Filter>
                                                |       <-      | |       ->      | |     Total     |    Relative    |   Duration   |
                                                | Frames  Bytes | | Frames  Bytes | | Frames  Bytes |      Start     |              |
10.10.1.4:1470             <-> 74.53.140.153:25                25 1,980 bytes      28 22 kB          53 24 kB        0.036986000         7.5777
================================================================================

┌──(kali㉿kali)-[~/SBT-DF203-Lab4]
└─$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info \
  | tee reports/smtp_packet_inventory.tsv

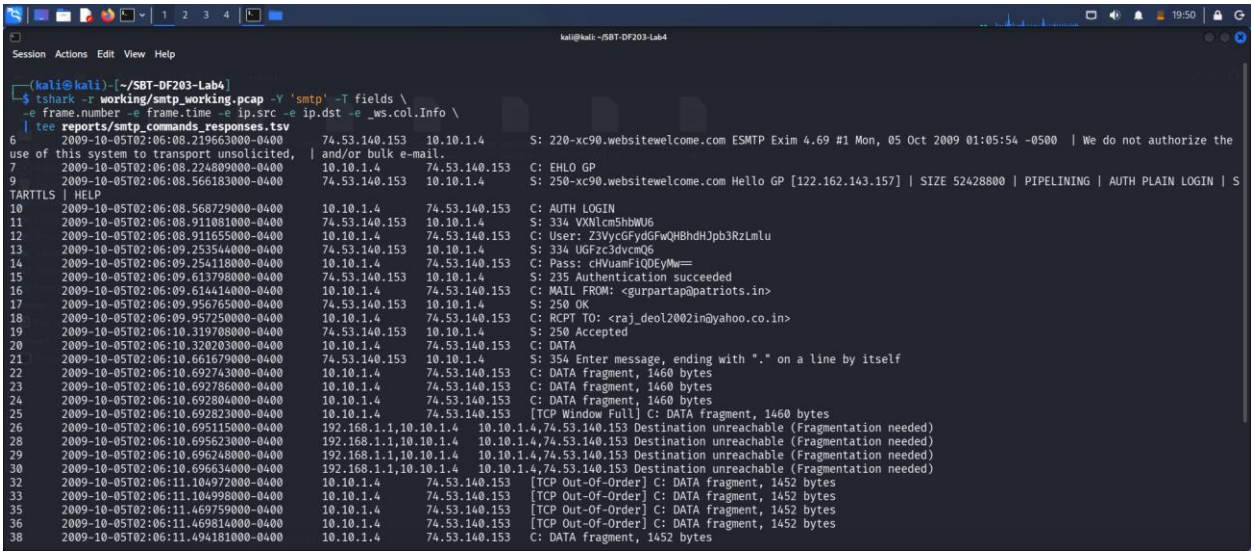
6       2009-10-05T02:06:08.219663000-0400      74.53.140.153   25      10.10.1.4       1470    S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500  | We do no
t authorize the use of this system to transport unsolicited,  | and/or bulk e-mail.
7       2009-10-05T02:06:08.224809000-0400      10.10.1.4       1470    74.53.140.153   25      C: EHLO GP
9       2009-10-05T02:06:08.566183000-0400      74.53.140.153   25      10.10.1.4       1470    S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] | SIZE 52428800 | PIPELINING | AUTH
 PLAIN LOGIN | STARTTLS | HELP
10      2009-10-05T02:06:08.568729000-0400      10.10.1.4       1470    74.53.140.153   25      C: AUTH LOGIN
11      2009-10-05T02:06:08.911081000-0400      74.53.140.153   25      10.10.1.4       1470    S: 334 VXNlcm5hbWU6
12      2009-10-05T02:06:08.911655000-0400      10.10.1.4       1470    74.53.140.153   25      C: User: Z3VycGFydGFwQHBhdHJpb3RzLmlu
13      2009-10-05T02:06:09.253544000-0400      74.53.140.153   25      10.10.1.4       1470    S: 334 UGFzc3dvcmQ6
14      2009-10-05T02:06:09.254118000-0400      10.10.1.4       1470    74.53.140.153   25      C: Pass: cHVuamFiQDEyMw==
15      2009-10-05T02:06:09.613798000-0400      74.53.140.153   25      10.10.1.4       1470    S: 235 Authentication succeeded
16      2009-10-05T02:06:09.614414000-0400      10.10.1.4       1470    74.53.140.153   25      C: MAIL FROM: <gurpartap@patriots.in>
17      2009-10-05T02:06:09.956765000-0400      74.53.140.153   25      10.10.1.4       1470    S: 250 OK
18      2009-10-05T02:06:09.957250000-0400      10.10.1.4       1470    74.53.140.153   25      C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19      2009-10-05T02:06:10.319708000-0400      74.53.140.153   25      10.10.1.4       1470    S: 250 Accepted
20      2009-10-05T02:06:10.320203000-0400      10.10.1.4       1470    74.53.140.153   25      C: DATA
21      2009-10-05T02:06:10.661679000-0400      74.53.140.153   25      10.10.1.4       1470    S: 354 Enter message, ending with "." on a line by itself
22      2009-10-05T02:06:10.692743000-0400      10.10.1.4       1470    74.53.140.153   25      C: DATA fragment, 1460 bytes
23      2009-10-05T02:06:10.692786000-0400      10.10.1.4       1470    74.53.140.153   25      C: DATA fragment, 1460 bytes
```

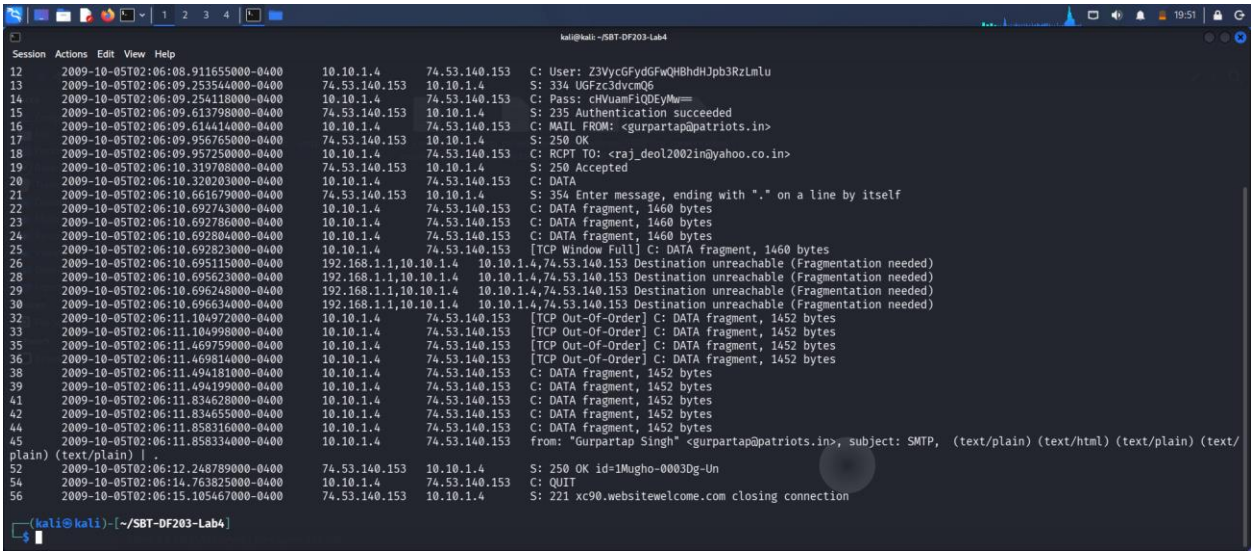
```
12      2009-10-05T02:06:08.911655000-0400      10.10.1.4       1470    74.53.140.153   25      C: User: Z3VycGFydGFwQHBhdHJpb3RzLmlu
13      2009-10-05T02:06:09.253544000-0400      74.53.140.153   25      10.10.1.4       1470    S: 334 UGFzc3dvcmQ6
14      2009-10-05T02:06:09.254118000-0400      10.10.1.4       1470    74.53.140.153   25      C: Pass: cHVuamFiQDEyMw==
15      2009-10-05T02:06:09.613798000-0400      74.53.140.153   25      10.10.1.4       1470    S: 235 Authentication succeeded
16      2009-10-05T02:06:09.614414000-0400      10.10.1.4       1470    74.53.140.153   25      C: MAIL FROM: <gurpartap@patriots.in>
17      2009-10-05T02:06:09.956765000-0400      74.53.140.153   25      10.10.1.4       1470    S: 250 OK
18      2009-10-05T02:06:09.957250000-0400      10.10.1.4       1470    74.53.140.153   25      C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19      2009-10-05T02:06:10.319708000-0400      74.53.140.153   25      10.10.1.4       1470    S: 250 Accepted
20      2009-10-05T02:06:10.320203000-0400      10.10.1.4       1470    74.53.140.153   25      C: DATA
21      2009-10-05T02:06:10.661679000-0400      74.53.140.153   25      10.10.1.4       1470    S: 354 Enter message, ending with "." on a line by itself
22      2009-10-05T02:06:10.692743000-0400      10.10.1.4       1470    74.53.140.153   25      C: DATA fragment, 1460 bytes
23      2009-10-05T02:06:10.692786000-0400      10.10.1.4       1470    74.53.140.153   25      C: DATA fragment, 1460 bytes
24      2009-10-05T02:06:10.692804000-0400      10.10.1.4       1470    74.53.140.153   25      C: DATA fragment, 1460 bytes
25      2009-10-05T02:06:10.692823000-0400      10.10.1.4       1470    74.53.140.153   25      [TCP Window Full] C: DATA fragment, 1460 bytes
26      2009-10-05T02:06:10.695115000-0400      192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140.153 25      Destination unreachable (Fragmentation needed)
28      2009-10-05T02:06:10.695623000-0400      192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140.153 25      Destination unreachable (Fragmentation needed)
29      2009-10-05T02:06:10.696248000-0400      192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140.153 25      Destination unreachable (Fragmentation needed)
30      2009-10-05T02:06:10.696634000-0400      192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140.153 25      Destination unreachable (Fragmentation needed)
32      2009-10-05T02:06:11.104972000-0400      10.10.1.4       1470    74.53.140.153   25      [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
33      2009-10-05T02:06:11.104998000-0400      10.10.1.4       1470    74.53.140.153   25      [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
35      2009-10-05T02:06:11.469759000-0400      10.10.1.4       1470    74.53.140.153   25      [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
36      2009-10-05T02:06:11.469814000-0400      10.10.1.4       1470    74.53.140.153   25      [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
38      2009-10-05T02:06:11.494181000-0400      10.10.1.4       1470    74.53.140.153   25      C: DATA fragment, 1452 bytes
39      2009-10-05T02:06:11.494199000-0400      10.10.1.4       1470    74.53.140.153   25      C: DATA fragment, 1452 bytes
41      2009-10-05T02:06:11.834628000-0400      10.10.1.4       1470    74.53.140.153   25      C: DATA fragment, 1452 bytes
42      2009-10-05T02:06:11.834655000-0400      10.10.1.4       1470    74.53.140.153   25      C: DATA fragment, 1452 bytes
44      2009-10-05T02:06:11.858316000-0400      10.10.1.4       1470    74.53.140.153   25      C: DATA fragment, 1452 bytes
45      2009-10-05T02:06:11.858334000-0400      10.10.1.4       1470    74.53.140.153   25      from: "Gurpartap Singh" <gurpartap@patriots.in>, subject: SMTP,  (text/plain) (text/html) (te
xt/plain) (text/plain) (text/plain) | .
52      2009-10-05T02:06:12.248789000-0400      74.53.140.153   25      10.10.1.4       1470    S: 250 OK id=1Mugho-0003Dg-Un
54      2009-10-05T02:06:14.763825000-0400      10.10.1.4       1470    74.53.140.153   25      C: QUIT
56      2009-10-05T02:06:15.105467000-0400      74.53.140.153   25      10.10.1.4       1470    S: 221 xc90.websitewelcome.com closing connection

┌──(kali㉿kali)-[~/SBT-DF203-Lab4]
└─$
```

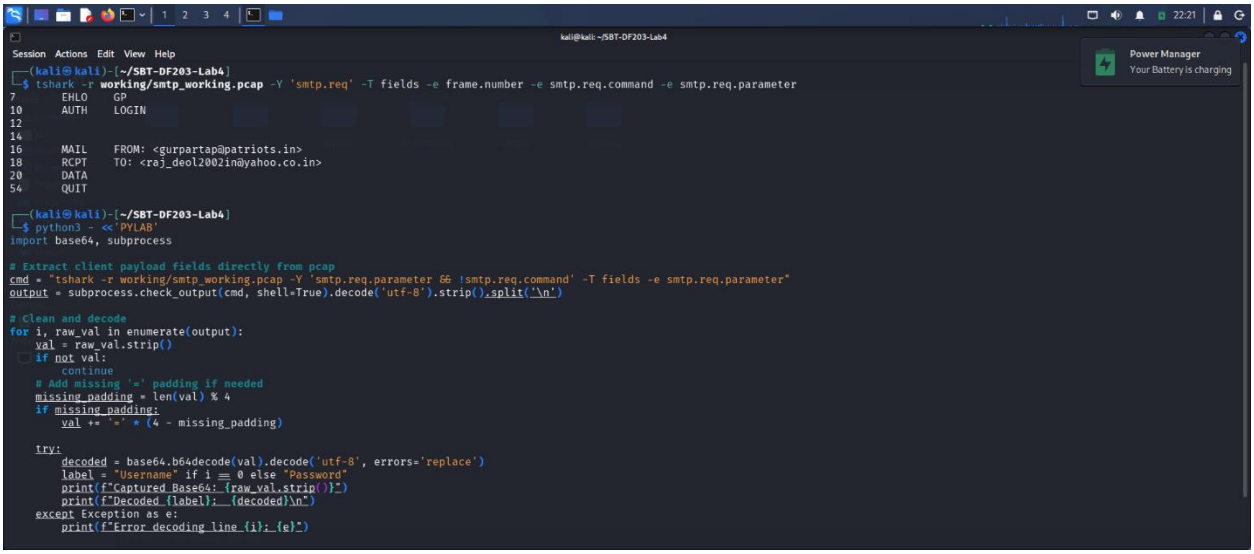
Part B - Identify Commands and Response Codes

Sequence	Direction	Command/code	Meaning	Timestamp
1	Server to client	220	Service ready	
2	Client to server	EHLO/HELO	Client greeting	
3	Client to server	AUTH	Authentication method	
4	Client to server	MAIL FROM	Envelope sender	
5	Client to server	RCPT TO	Envelope recipient	
6	Client to server	DATA	Message content begins	
7	Client to server	QUIT	Session termination	



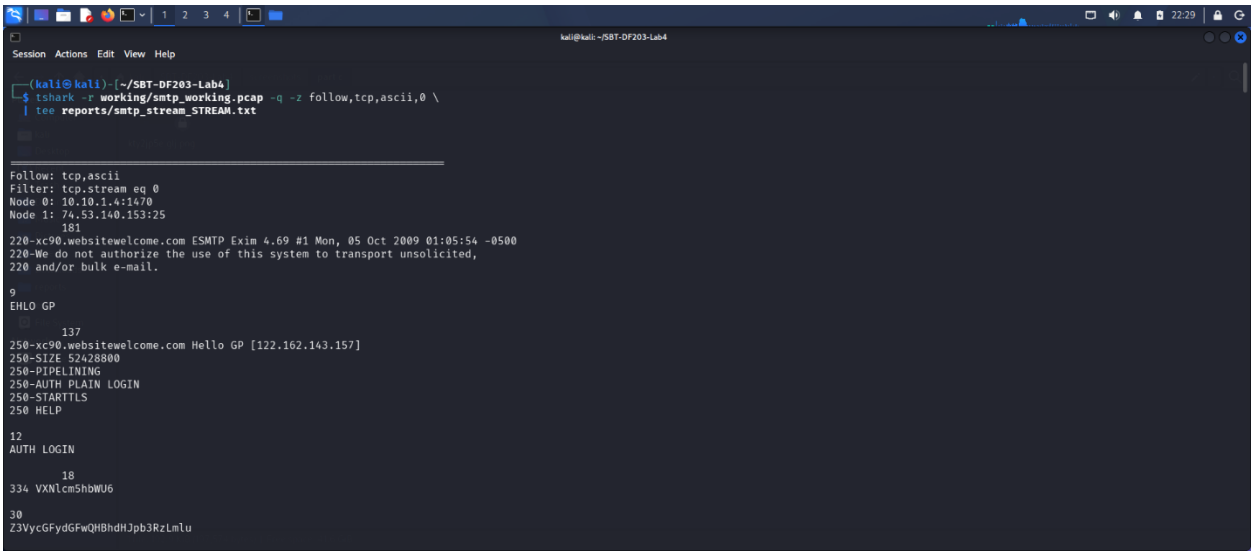


Part C - Decode Base64 Authentication Evidence



Part D - Reconstruct the Email Message





```
Session  Actions  Edit  View  Help

————=_NextPart_000_00
29
04_01CA45B0.095693F0--

.

        28
250 OK id=1Mugho-0003Dg-Un

6
QUIT

        48
221 xc90.websitewelcome.com closing connection

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┌──(kali㉿kali)-[~/SBT-DF203-Lab4]
└─$ grep -Ei '^(Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):' \
  reports/smtp_stream_STREAM.txt | tee reports/message_headers.txt
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;

┌──(kali㉿kali)-[~/SBT-DF203-Lab4]
└─$
```

Part E - Determine Client, Hosts and Network Metadata

```
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~/SBT-DF203-Lab4]
└─$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
  -e ip.dst -e tcp.dstport -e tcp.stream \
  | tee reports/smtp_network_metadata.tsv

6       1254722768.219663000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   25      10.10.1.4       1470    0
7       1254722768.224809000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
9       1254722768.566183000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   25      10.10.1.4       1470    0
10      1254722768.568729000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
11      1254722768.911081000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   25      10.10.1.4       1470    0
12      1254722768.911655000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
13      1254722769.253544000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   25      10.10.1.4       1470    0
14      1254722769.254118000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
15      1254722769.613798000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   25      10.10.1.4       1470    0
16      1254722769.614414000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
17      1254722769.956765000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   25      10.10.1.4       1470    0
18      1254722769.957250000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
19      1254722770.319708000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   25      10.10.1.4       1470    0
20      1254722770.320203000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
21      1254722770.661679000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   25      10.10.1.4       1470    0
22      1254722770.692743000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
23      1254722770.692786000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
24      1254722770.692804000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
25      1254722770.692823000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
26      1254722770.695115000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140.153 25      0
28      1254722770.695623000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140.153 25      0
29      1254722770.696248000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140.153 25      0
30      1254722770.696634000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140.153 25      0
32      1254722771.104972000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
33      1254722771.104998000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
35      1254722771.469759000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
36      1254722771.469814000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
38      1254722771.494181000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
```

```
Session  Actions  Edit  View  Help
18      1254722769.957250000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
19      1254722770.319708000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   25      10.10.1.4       1470    0
20      1254722770.320203000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
21      1254722770.661679000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   25      10.10.1.4       1470    0
22      1254722770.692743000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
23      1254722770.692786000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
24      1254722770.692804000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
25      1254722770.692823000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
26      1254722770.695115000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140.153 25      0
28      1254722770.695623000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140.153 25      0
29      1254722770.696248000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140.153 25      0
30      1254722770.696634000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140.153 25      0
32      1254722771.104972000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
33      1254722771.104998000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
35      1254722771.469759000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
36      1254722771.469814000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
38      1254722771.494181000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
39      1254722771.494199000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
41      1254722771.834628000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
42      1254722771.834655000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
44      1254722771.858316000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
45      1254722771.858334000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
52      1254722772.248789000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   25      10.10.1.4       1470    0
54      1254722774.763825000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       1470    74.53.140.153   25      0
56      1254722775.105467000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   25      10.10.1.4       1470    0

┌──(kali㉿kali)-[~/SBT-DF203-Lab4]
└─$ tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
  -T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
  | tee reports/client_indicators.tsv

22      0               Line-based text data (47 lines)
26      0               Line-based text data (12 lines)

┌──(kali㉿kali)-[~/SBT-DF203-Lab4]
└─$
```

Part F - Encryption and Evidential Limitations

STARTTLS Request Verification

In Frame 9, the server advertised STARTTLS support as part of its 250 response line. The client did not issue a STARTTLS request. It proceeded directly to AUTH LOGIN (Frame 10) over cleartext TCP. All subsequent command strings, challenge responses, and email content remained plain text throughout the capture.

Forensic Significance of Port Numbers

Port numbers alone do not prove encryption. Operating on TCP port 25 or 587 does not guarantee plain text or TLS; a session on port 25 can upgrade to TLS via STARTTLS, while traffic on port 465 (SMTPS) could theoretically misconfigure. Protocol dissection and packet-level inspection are required to verify whether a TLS handshake (Client Hello / Server Hello) occurred.

Evidential Limitations of Encrypted Traffic

Had STARTTLS been successfully negotiated:

1. Readable Metadata: Endpoint IP addresses, MAC addresses, TCP ports, session timing, frame lengths, and packet byte counts would remain visible.
2. Hidden Payload: Authentication credentials, envelope headers (MAIL FROM/RCPT TO), and email body content would be completely unreadable without access to private decryption keys or session secrets.

Question	Finding
When did the SMTP session start and end?	2009-10-05 02:06:08.219663 to 02:06:15.105467 EDT
Client IP/MAC and port	IP: 10.10.1.4 | Ephemeral Port: Transmitted over active TCP Stream
Server IP/MAC and port	IP: 74.53.140.153 | Port: 25
SMTP server banner	220-xc90.websitewelcome.com ESMTP Exim 4.69 #1
Client software	GP (NAT IP: 122.162.143.157)
Authentication method	AUTH LOGIN (Base64 encoded)
Envelope sender and recipient	Sender: gurpartap@patriots.in | Recipient: raj_deol2002in@yahoo.co.in
Message From/To/Subject	From: "Gurpartap Singh" <gurpartap@patriots.in> | To: raj_deol2002in@yahoo.co.in | Subject: SMTP
Message body type	Multipart (text/plain and text/html)
Attachment present?	No
STARTTLS/TLS observed?	No (Advertised by server, but unused by client)
Key limitations	Absence of TLS allowed complete payload and credential reconstruction.